

A Computational Model of Allophony as Dispersion over Phonetic Sequences

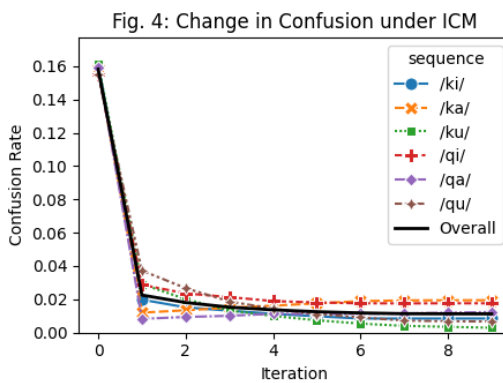
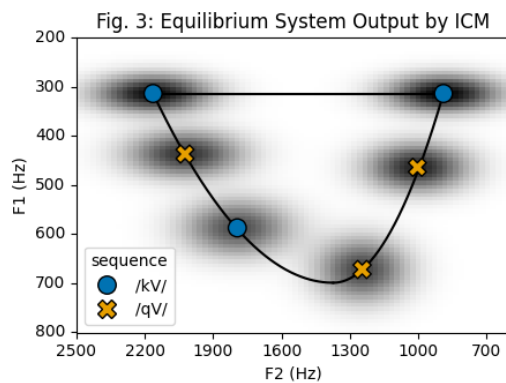
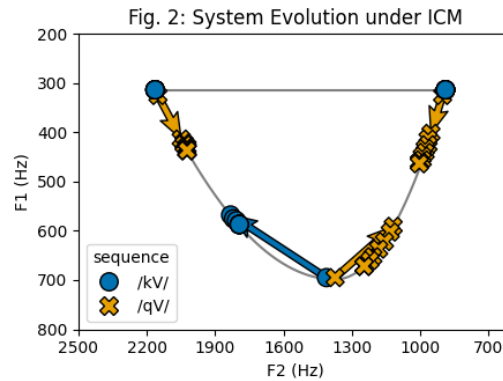
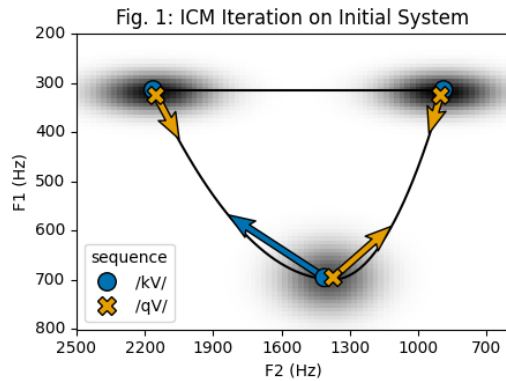
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Accepted for an oral presentation at LSA 2026

Dispersion Theory (Liljencrantz & Lindblom, 1972; Schwartz et al., 1997; Flemming, 2004) contends that the world's most common sound systems reflect a pressure to maximize perceptual contrast, e.g. the cross-linguistically ubiquitous 'corner' vowels /iau/ minimize risk of listener misperception because they are maximally acoustically distinct. Although past dispersion models have predicted phoneme inventories by optimizing over sets of individual phonological segments, decades of experimental research have shown that perception of speech sounds is sensitive to larger units such as words (Ganong, 1980) and phone sequences (Massaro & Cohen, 1983; Steffman & Sundara, 2024). Given such results, I propose that a perceptually optimal sound system should optimize not only segments but also multi-segmental patterns. In this vein, the current work generalizes the notion of perceptual contrast maximization to biphone sequences using a computational sound change model called Iterated Confusion Minimization. This model is shown to predict certain patterns of phonologically conditioned allophony, exemplified here with the typologically common lowering/retraction of vowels adjacent to uvulars, cf. Arabic (McCarthy, 1994), West Greenlandic (Rischel, 1974), and Quechua (Gallagher, 2016).

Iterated Confusion Minimization (ICM) simulates the interaction of mathematically optimal listeners and speakers over time. Like existing Bayesian models of optimal listeners (Clayards et al., 2008; Feldman et al., 2009; Kleinschmidt, 2019), ICM models phonetic categories as probability distributions, allowing for a mathematically precise measure of listener confusion based on distribution overlap. ICM carries out local optimization over this confusion measure, tending to increase phonetic dispersion over time. Forthcoming work (under review, omitted for anonymity) has shown that ICM predicts common sound change phenomena like chain shifts (Labov, 1994) and outperforms past dispersion models (Schwartz et al., 1997; de Boer, 1999) in explaining both typologically dominant vowel systems and typological variation. Extending ICM to model allophony suggests a unified explanation of these varied phenomena as rooted in the same basic mechanism, i.e. maximization of perceptual contrast.

The ICM simulation of the figures below demonstrates an allophonic split of the vowels /iau/ into [iæu] after /k/ and [eao] after /q/. The sequences {/ki/, /ka/, /ku/, /qi/, /qa/, /qu/} are modeled as three-dimensional normal distributions based on an empirically informed model of possible vowel distributions over a two-dimensional formant space combined with idealized overlapping one-dimensional /k/ and /q/ distributions ($N(-1,1)$ and $N(1,1)$, not displayed due to limited visualizable dimensions). Figure 1 shows that an initial ICM iteration over a system without significant allophony immediately amplifies small differences in vowel quality attributable to coarticulation with /q/ into a robust allophonic split. The system continues to undergo small adjustments (Figure 2) until an equilibrium system {[ki], [kæ], [ku], [qe], [qɑ], [qo]} emerges (Figure 3). ICM predicts this result because the availability of vowel allophones as a perceptual cue for the highly confusable /k/-/q/ contrast (cf. Gallagher, 2016) reduces confusion (Figure 4). Additional simulations over systems including more vowel categories with greater overlap do not yield such allophony, a prediction which will be tested against typology in future work.



References:

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